LexiPic Suite

Overview & Cross-Module Architecture Documentation

Modules covered: LexiPic v1.2.2 · LexiPic Kiosk v1.2.1 · LexiPic Kiosk PWA v1.0.0

Publisher: BhoomiTech Heritage and Development Foundation

Document type: Suite-level overview (see companion technical documents for module detail)

Status: Release documentation

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1. Introduction

LexiPic is a suite of three companion software products built to record, archive, and display the vocabulary of heritage and under-resourced languages — currently Bishnupriya Manipuri, Assamese, and Bengali, with support for adding further languages. The suite spans content authoring inside WordPress, multi-kiosk distribution and management, and a server-less kiosk application that can run on dedicated hardware with no live network connection.

This document describes the suite as a whole: what each module is responsible for, how data moves between them, the deployment patterns they support together, and the security model that spans all three. It is intended as the entry point for anyone evaluating, installing, or maintaining the LexiPic product family. Each module additionally has its own detailed technical document (file structure, database schema, REST API reference, and front-end architecture) — this overview cross-references those documents rather than duplicating their detail.

> **Companion documents:** LexiPic (Core Plugin) — Technical Documentation · LexiPic Kiosk (Plugin) — Technical Documentation · LexiPic Kiosk PWA — Technical Documentation.

2. The Suite at a Glance

The three modules are independently installable, but designed to interoperate through a shared, self-contained JSON export format. Each can also be used on its own.

Module	Version	Type	Runs on	Primary role
LexiPic	1.2.2	WordPress plugin	A WordPress site	Content authoring: capture words (photo, audio, native script, transliteration), organize them into Sets, and produce the JSON export that feeds the other two modules.
LexiPic Kiosk	1.2.1	WordPress plugin	A WordPress site (same or different from LexiPic)	Distribution and central management of word sets across one or more touchscreen kiosks, with its own storage, REST API, and PIN-protected on-screen admin panel.
LexiPic Kiosk PWA	1.0.0	Static web app (PWA)	A kiosk device's browser, online or fully offline	The actual kiosk display application. Either shows a remote LexiPic Kiosk web page in an iframe, or runs entirely offline from word-set JSON files loaded from a USB drive.

2.1 Why three separate modules

- Separation of authoring from display. LexiPic (Core) is built for the people recording words — it needs microphone capture, an on-screen keyboard-like transliteration input, and moderation/access control. A public kiosk has none of those needs.
- Separation of one WordPress site from many physical kiosks. A single organization may run one WordPress site for content creation, but deploy a dozen physical kiosk machines across different locations — those kiosks need centralized management without giving every kiosk operator a WordPress login.
- Offline resilience for physical hardware. A kiosk in a museum or community hall cannot depend on uninterrupted internet access. The PWA module exists specifically so a kiosk can run from local files with zero server dependency, while still being centrally distributable when a network is available.

3. The Three Modules

3.1 LexiPic (Core Plugin) v1.2.2

A WordPress plugin (text domain lexipic) that turns any WordPress page into a structured "picture dictionary" archive. Contributors pair a photograph with a spoken recording, the word in native script, and a Roman transliteration, generated live in the browser using a built-in phonetic input method engine ("CompLing AI"). Entries are grouped into Sets, each with its own language. The plugin exposes a full lexipic/v1 REST namespace and a [lexipic] shortcode front end with three tabs (Archive, Add Entry, Sets), gated by WordPress login state and a configurable per-tab username allowlist.

Images are stored as binary blobs directly in the plugin's own database table; audio recordings are stored as files under wp-content/uploads/lexipic/ and, where FFmpeg is available on the server, are converted from .webm to .mp3 in the background a few seconds after upload. A full JSON import/export feature makes every Set portable as a single self-contained file (photos and audio embedded as base64), which is the same format the Kiosk plugin and the PWA both consume.

See the companion document "LexiPic (Core Plugin) — Technical Documentation" for the database schema, full REST endpoint reference, the IME/transliteration engine architecture, and the language-registry/file-management system under Settings → IME Engine.

3.2 LexiPic Kiosk (Plugin) v1.2.1

A second, independent WordPress plugin (text domain lexipic-kiosk) purpose-built for displaying word sets on touchscreen kiosks and for managing those kiosks centrally. It maintains its own database tables (entirely separate from LexiPic's and from the WordPress Media Library), its own lexipic-kiosk/v1 REST namespace, and a PIN-gated on-screen admin panel so a kiosk operator can import, rename, reorder, activate/deactivate, or delete sets directly from the kiosk's touchscreen without a WordPress login.

A kiosk can be embedded inline on any page via the [lexipic_kiosk] shortcode, or displayed as a dedicated fullscreen page at a clean kiosk URL with no theme header, footer, or admin toolbar. The plugin also publishes a domain allowlist (LexiPic Kiosk → Approved Kiosk URLs) that the standalone PWA module consults before it is allowed to display any "Remote URL" kiosk page — this is the integration point between the two plugin-side modules and the PWA.

See the companion document "LexiPic Kiosk (Plugin) — Technical Documentation" for the database schema (including the 1.0.0 → 1.1.0 media-storage migration), the full REST endpoint reference, the PIN/session security model, and the import-normalizer that accepts both this suite's current export format and the legacy flat-array format from the original standalone prototype.

3.3 LexiPic Kiosk PWA v1.0.0

A dependency-free, installable Progressive Web App — plain HTML/CSS/JavaScript with no build step and no server component ("No server. No PHP. Just open index.html.", per the app's own source comments). It is the actual software that typically runs full-screen on the kiosk hardware itself.

On first launch it offers a choice of two source modes, remembered across reloads:

- **Remote URL mode** — displays a remote kiosk web page (e.g. a LexiPic Kiosk plugin fullscreen URL) directly in a fullscreen iframe, with automatic reload-on-failure. The destination domain must appear on the WordPress site's Approved Kiosk URLs allowlist, and changing an already-configured source requires the same PIN configured in LexiPic Kiosk → Settings.

- **USB Drive mode** — the operator selects a local lexipic-kiosk-data folder (via the File System Access API). The app then renders its own tile grid and card slider entirely client-side from a settings.json manifest and one JSON file per word set — no network connection required at any point, including on subsequent boots.

A service worker pre-caches the entire app shell on first load so the application itself always starts even with no connectivity; in USB mode this means a kiosk can run indefinitely fully offline. See the companion document "LexiPic Kiosk PWA — Technical Documentation" for the security model, IndexedDB schema, and the settings.json / per-set JSON file formats.

4. How Data Flows Through the Suite

All three modules speak a common JSON "export" vocabulary for a word Set. This is what makes the suite composable: any module that can produce this shape can feed any module that consumes it.

4.1 The canonical Set export format

LexiPic's REST export endpoint (GET /lexipic/v1/sets/{id}/export) produces the reference shape. Every entry's image and audio are embedded as base64 data URIs, so a single .json file is fully self-contained and portable between sites with no separate media transfer:

```
{
"lexipic_version": "1.2.2",
"set": {
"name": "Domestic Animals",
"slug": "domestic-animals",
"language": "bpm"
  },
"entries": [
    {
"id": 7,
"word_script": "<word in native script>",
"word_roman": "guru",
"description": "Optional note about the entry.",
"image": "data:image/jpeg;base64,...",
"audio": "data:audio/webm;base64,...",
"image_mime": "image/jpeg",
"audio_mime": "audio/webm"
    }
  ]
}
```

Both downstream modules (Kiosk plugin and PWA) accept this exact shape, and both also accept the legacy flat-array format produced by the original, pre-suite standalone prototype (an array of entry objects with no set wrapper, using the older field names heritage / transliteration) — see §4.5.

4.2 Path A — WordPress-only distribution

1. A contributor records entries in LexiPic's Add Entry tab on the WordPress site; CompLing AI handles live script transliteration as they type.
2. An editor exports the finished Set as JSON from LexiPic's Import/Export admin screen, or directly via the REST export endpoint.
3. The same or a different WordPress site administrator imports that JSON file into LexiPic Kiosk → Word Sets. The Kiosk plugin's import normalizer converts every embedded base64 image/audio into a real file under wp-content/uploads/lexipic-kiosk/, so the database only ever stores short URLs.
4. The kiosk is displayed either inline via [lexipic_kiosk] or at its dedicated fullscreen URL, and is centrally managed (renamed, reordered, activated/deactivated) from LexiPic Kiosk → Word Sets going forward.

4.3 Path B — Offline kiosk hardware (USB)

5. A Set is exported from LexiPic (or from LexiPic Kiosk's own /backup endpoint) as JSON, exactly as in Path A.
6. The exported file is placed inside a lexipic-kiosk-data folder on a USB drive, alongside a settings.json manifest listing each set's slug, display name, enabled flag, and group label (and optionally the kiosk's idle-timer/autoplay defaults).
7. On the kiosk device, the LexiPic Kiosk PWA is launched and "USB Drive" mode is selected; the operator picks that folder once. The PWA reads settings.json and the listed per-set JSON files directly from disk and renders the tile grid / card slider entirely client-side.
8. No WordPress site needs to be reachable at any point after the initial export — this path is fully offline-capable indefinitely.

4.4 Path C — Remote URL kiosk (PWA as a managed browser)

Here the PWA does not parse any JSON at all. Instead, in "Remote URL" mode it simply displays a LexiPic Kiosk fullscreen page inside a fullscreen iframe, functioning as a locked-down kiosk browser. The actual data lives and is served entirely from the WordPress site running LexiPic Kiosk (Path A). This mode requires an active network connection and exists to combine the PWA's kiosk chrome (auto-reload on connection loss, no browser address bar, app-like install) with centrally-managed WordPress content, without needing the operator to keep any local files in sync.

> **Security note:** Path C is gated by the destination WordPress site's Approved Kiosk URLs allowlist and, when changing an already-configured source, by that same site's kiosk PIN — see §6.

4.5 Legacy format compatibility

Both LexiPic's own import handler and LexiPic Kiosk's import normalizer also accept the flat-array JSON format produced by the project's original, pre-plugin standalone prototype application (a bare JSON array of entries, with field names heritage and transliteration instead of word_script and word_roman). This means archives created before the current suite existed can still be brought forward into either plugin without manual conversion.

5. Deployment Scenarios

Because the modules are independently installable, an organization can adopt only as much of the suite as it needs.

Scenario	Modules used	Notes
Content archive only — a browsable language archive embedded in a WordPress site, no physical kiosk	LexiPic only	The [lexipic] shortcode’s Archive tab already provides swipeable browsing; no kiosk hardware needed.
Single in-house kiosk on the same WordPress site	LexiPic + LexiPic Kiosk	LexiPic for authoring, LexiPic Kiosk’s shortcode or fullscreen URL for the public-facing touchscreen display.
Multiple kiosks at different physical locations, all online	LexiPic + LexiPic Kiosk (+ PWA in Remote URL mode, optional)	Every kiosk points at the same LexiPic Kiosk fullscreen URL (directly in a browser, or via the PWA for kiosk-mode chrome); content is managed once, centrally.
Kiosk with no reliable network at its physical location	LexiPic (export only) + LexiPic Kiosk PWA in USB mode	No live WordPress connection is required at the kiosk once the USB drive is prepared; LexiPic Kiosk plugin is optional in this scenario.
Mixed fleet — some kiosks online, some offline	All three, used per-device	Each physical kiosk independently chooses Remote URL or USB mode in the PWA; both modes are re-selectable at any time from that device’s setup screen.

6. Cross-Module Security Model

Each module layers its own access control, and the PWA deliberately reuses the Kiosk plugin's existing credential rather than introducing a second, weaker one.

6.1 LexiPic (Core Plugin)

- Viewing the archive requires being logged in to WordPress, unless the site administrator enables the "Public Archive" setting.
- Adding entries and managing Sets each require either WordPress Administrator status or explicit per-tab username allowlisting under LexiPic → Settings.
- Every REST write additionally requires a logged-in WordPress session, independent of what the front-end UI shows or hides.

6.2 LexiPic Kiosk (Plugin)

- A dedicated manage_lexipic_kiosk WordPress capability (granted to Administrators automatically; assignable to other roles) gates wp-admin management screens.
- A separate kiosk PIN (hashed with `wp_hash_password()`, never transmitted or stored in plain form after being set) gates the on-screen kiosk admin panel, independent of any WordPress login. A successful PIN check returns a short-lived (30-minute) session token used for subsequent REST writes from that device.
- PIN verification is rate-limited (5 attempts per minute per IP address) to resist brute-forcing — unlike the original prototype's hardcoded client-side password check.
- Site-wide kiosk defaults (idle timer, autoplay) and the Approved Kiosk URLs allowlist both require an actual logged-in WordPress session with the management capability — a kiosk PIN alone is not sufficient for either, since both affect every kiosk, not just the device the PIN was entered on.
- The dedicated fullscreen kiosk URL explicitly suppresses the WordPress admin toolbar, even for an administrator who happens to still be logged in on that browser, so a public kiosk screen can never expose a one-click path into wp-admin.

6.3 LexiPic Kiosk PWA

- Any URL saved in Remote URL mode is checked, by hostname, against the domain allowlist published by the WordPress site's LexiPic Kiosk → Approved Kiosk URLs page. The check fails closed: if the allowlist endpoint cannot be reached and nothing was cached from a previous successful check, the save is blocked rather than allowed through unchecked.
- Changing an already-configured source (URL or USB) requires the same kiosk PIN configured in the Kiosk plugin's Settings screen, verified by calling that plugin's existing /pin/verify endpoint — there is no second, PWA-specific PIN to manage. First-ever setup on a freshly-deployed kiosk is not PIN-gated, since there is nothing yet to protect.
- A saved Remote URL is re-validated against the live allowlist on every boot and periodically while running; an administrator revoking a domain from wp-admin disconnects any kiosk currently displaying it without needing physical access to that device.
- The kiosk session token is held only in memory for the lifetime of the page (never written to localStorage, sessionStorage, or a cookie), so a shared or idle kiosk cannot leak a still-valid admin session across a reload or to another browser tab.

7. Versioning & Compatibility

Module	Current version	Minimum WordPress	Minimum PHP	Format compatibility
LexiPic	1.2.2	5.8	7.4	Produces the canonical export format (§4.1); accepts both that format and the legacy flat-array PWA prototype format on import.
LexiPic Kiosk	1.2.1	6.0	7.4	Accepts the canonical export format and the legacy flat-array format via its import normalizer; its own /backup format is versioned independently (backupFormatVersion: 1).
LexiPic Kiosk PWA	1.0.0	n/a (browser-based)	n/a	USB mode reads the canonical per-set JSON format plus a settings.json manifest; Remote URL mode is format-agnostic (it displays a page, not data).

> **Browser requirement:** USB mode in the PWA requires the File System Access API (showDirectoryPicker), currently supported in Chromium-based browsers (Chrome, Edge) version 86 and later. Remote URL mode has no such requirement.

8. Glossary

Set A named, single-language collection of word entries (e.g. "Domestic Animals"). The basic unit of import/export across the suite.

Entry One word/card within a Set: native-script word, Roman transliteration, optional description, one photo, one audio recording.

IME / CompLing AI engine LexiPic's built-in phonetic input method: a fixed, deterministic transliteration engine (LexiPic Core only) that converts Roman keystrokes to native script live, and vice versa, driven by per-language JSON map files.

Allowlist (Approved Kiosk URLs) A list of domains, published by LexiPic Kiosk, that the PWA is permitted to display in Remote URL mode.

Kiosk PIN A short numeric/alphanumeric code, set once in LexiPic Kiosk → Settings, that unlocks the on-screen kiosk admin panel (both the plugin's own embedded kiosk and the PWA's "change source" action reuse this same PIN).

Remote URL mode A PWA source mode that displays a live WordPress-hosted kiosk page in a fullscreen iframe.

USB mode A PWA source mode that reads word-set data from a local folder, with no network dependency.

Fullscreen kiosk URL A dedicated, theme-free WordPress page (e.g. /kiosk/) published by LexiPic Kiosk specifically for full-screen device display.

App shell The PWA's own HTML/CSS/JS/icons, pre-cached by its service worker so the application itself loads even with no network connection.

9. Where to Go Next

This overview intentionally stays at the suite level. For implementation-level detail, consult the companion technical document for the relevant module:

Document	Covers
LexiPic (Core Plugin) — Technical Documentation	Database schema, full REST API reference, IME/transliteration engine internals, language registry & file management, audio conversion, front-end app architecture.
LexiPic Kiosk (Plugin) — Technical Documentation	Database schema & migration history, full REST API reference, PIN/session security, media storage strategy, import normalizer, admin screens, front-end caching architecture.
LexiPic Kiosk PWA — Technical Documentation	Source-mode architecture, service worker & offline strategy, IndexedDB schema, allowlist/PIN security flow, settings.json and per-set JSON formats, deployment to kiosk hardware.